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INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

ER Diagram for an Online Library System

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Technical Presentation



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Introduction

What is an Online Library Management System?

- An Online Library Management System is software that helps manage:
- Books
- Members
- Librarians
- Book Borrowing
- Book Returning
- Reservations
- Fine Collection

Objectives

- Store library information efficiently
- Reduce manual work
- Improve book searching
- Maintain borrowing records



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What is an ER Diagram?

- An **Entity Relationship (ER) Diagram** is a graphical representation of a database.
- It shows:
- Entities
- Attributes
- Relationships
- Primary Keys
- Foreign Keys

Advantages

- Easy database design
- Reduces redundancy
- Improves data integrity
- Helps developers understand the database

Main Components of an ER Diagram

Entity & Attribute	Relationship & Keys
Entity	Relationship
A real-world object in the database.	Shows how entities are connected.
Examples:	Examples:
• Book	• Member → Borrows → Book
• Member	• Librarian → Manages → Book
• Librarian	
Attribute	Keys
Describes the properties of an entity.	Primary Key: Uniquely identifies each record.
Example (Book):	Example: Book_ID
• Book_ID	Foreign Key: Connects related entities.
• Title	Example: Member_ID (in Borrow table)
• Author	
• Publisher	



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Main Entity

- The Online Library contains the following entities:
- Member
- Book
- Author
- Publisher
- Librarian
- Borrow
- Reservation
- Fine



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ER Diagram



Real-World Applications

1. Online Library Management

- Manage books, members, borrowing, and returns.

2. Banking System

- Store customer, account, loan, and transaction details.

3. Hospital Management

- Manage patients, doctors, appointments, and medical records.

4. E-Commerce

- Handle products, customers, orders, and payments.

5. University Management

- Store student, faculty, course, and examination information.

6. Airline Reservation System

- Manage flights, passengers, bookings, and ticket details.

Example Use Case

- Student: Saran
- Book: Database System Concepts
- Issue Date:
24 July
- Return Date:
30 July
- Fine:
₹180 (Late Return)

```
graph TD; Student --> SearchBook[Search Book]; SearchBook --> IssueRequest[Issue Request]; IssueRequest --> LibrarianApproval[Librarian Approval]; LibrarianApproval --> BookIssued[Book Issued]; BookIssued --> ReturnBook[Return Book]; ReturnBook --> FineCalculation[Fine Calculation (if delayed)];
```


Conclusion

- The **ER Diagram for the Online Library System** provides a clear structure for managing books, members, librarians, loans, reservations, and fines.
- It defines the relationships between entities, ensuring accurate data storage, reducing redundancy, and maintaining data integrity.
- The ER diagram serves as the foundation for designing an efficient, reliable, and scalable database, making library operations such as book management, borrowing, and returning more organized and effective.



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Future Scope

Future Enhancements

- E-book and Digital Library Support for reading books online.
- Mobile Application for easy access to library services anytime.
- AI-Based Book Recommendations based on users' reading history.
- Online Fine Payment through digital payment gateways.
- Barcode/RFID Integration for faster book issue and return.



THANK YOU....